

- 1 The p.d. across a resistor is 12 V. The power dissipated is 6.0 W.

Which statement is correct?

- A The charge passing through the resistor in one second is 2.0 coulomb.
- B The resistor transfers 6.0 joule for each coulomb passing through the resistor.
- C The resistor transfers 12 joule in 2.0 second.
- D The resistor dissipates 6.0 joule when the current is 2.0 ampere.

Your answer ☐

[1]

- 2 A copper wire is connected across a cell. The conduction electrons within the copper wire move.

Which statement is correct about these electrons?

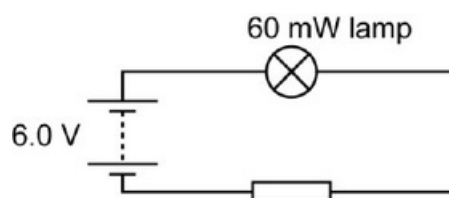
- A They drift towards the negative end of the cell.
- B They have random speeds because of collisions with other electrons.
- C They travel through the wire at the speed of light.
- D They collide with vibrating copper ions.

Your answer ☐

[1]

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3(a) A battery is connected in series with a lamp and a resistor as shown.



The battery has e.m.f. 6.0 V and negligible internal resistance. The potential difference across the lamp is 2.4 V and it dissipates 60 mW. The resistor has cross-sectional area of 2.0 mm². The number density of charge carriers (free electrons) within the resistor is $1.4 \times 10^{25} \text{ m}^{-3}$.

Calculate the resistance R of the resistor.

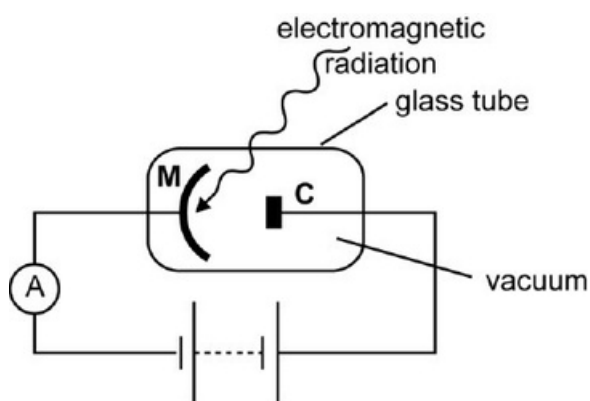
$R = \text{-----} \Omega$ [3]

(b) Calculate the mean drift velocity v of the free electrons within the resistor.

$v = \text{-----} \text{ m s}^{-1}$

(c) The number density of the free electrons in the connecting wires is greater than that of the resistor. The connecting wires have the same diameter as the resistor. State and explain whether the mean drift velocity of the free electrons would be smaller, the same, or larger than your value in (b).

 ----- [2]

**Fig. 19**

When the metal M is exposed to electromagnetic radiation, photoelectrons are ejected from the surface of the metal. These photoelectrons are collected at the electrode C and the sensitive ammeter indicates the presence of a tiny current.

The work function of the metal M is 2.3 eV.

The incident electromagnetic radiation has wavelength 5.1×10^{-7} m.

The ammeter reading is $0.24 \mu\text{A}$.

Calculate the number of photoelectrons reaching C in a time of 5.0 s.

number =[3]

- 5 A small heater is connected to a power supply. The power supply is switched on for 100 s. The current in the heater is 3.0 A and it dissipates 1200 J of thermal energy. What is the potential difference across the heater?
- A 0.25V
B 4.0V
C 12V
D 300V

Your answer

[1]



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- 6 A filament lamp X is part of an electrical circuit. The circuit has a battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance. The potential difference across the lamp can be increased continuously from 0 to 6.0 V. This potential difference is measured using a voltmeter.
The lamp glows brightly at 6.0 V.

(i) Draw a circuit diagram for this electrical arrangement.

[2]

- (ii) Describe and explain the variation of the resistance of this lamp as the potential difference across it is changed from 0 to 6.0 V.

EXAM PAPERS PRACTICE [4]

- (iii) The filament lamp X is now connected in a different circuit as shown in Fig. 16.

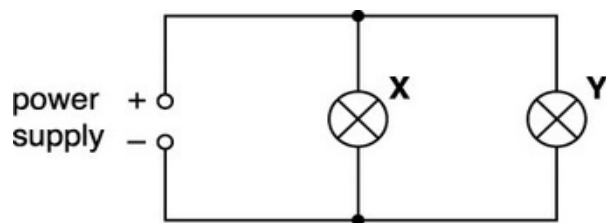


Fig. 16

The power dissipated in X is three times more than the power dissipated in the filament lamp Y. The filament wire of lamp X has a diameter half that of lamp Y.

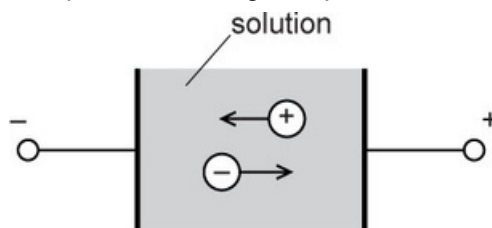
The filament wires of X and Y are made of the same material and are at the same temperature.

Calculate the ratio

$$\frac{\text{mean drift velocity of charge carriers in lamp X}}{\text{mean drift velocity of charge carriers in lamp Y}}$$

ratio = [3]

- 7 The diagram below shows the motion of positive and negative particles in a conducting solution.



Which statement is correct?

- A The current in the solution is zero.
- B The conventional current is to the left.
- C The positive particles are always protons.
- D The negative particles are always electrons.

Your answer

[1]

- 8 One million electrons travel between two points in a circuit.
The total energy gained by the electrons is 1.6×10^{-10} J.

What is the potential difference between the two points?

- A 1.6×10^{-16}
 - B V 1.6×10^{-4}
 - C V 1.0×10^3
 - D V 1.0×10^9
- V

Your answer

[1]

- 9 A small thin rectangular slice of semiconducting material has width a and thickness b and carries a current I . The current is due to the movement of electrons. Each electron has charge $-e$ and mean drift velocity v . A uniform magnetic field of flux density B is perpendicular to the direction of the current and the top face of the slice as shown in Fig. 2.1.

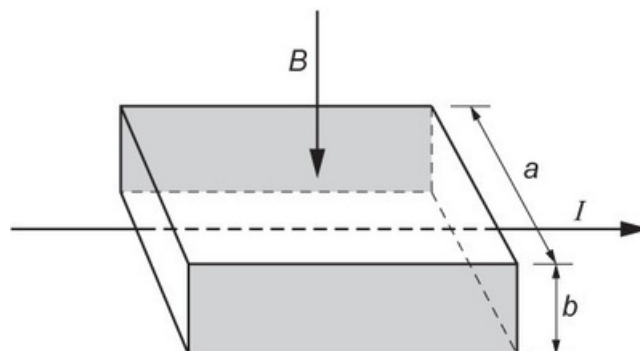


Fig. 2.1

Here are some data for the slice in a particular experiment.

number of conducting electrons per cubic metre, $n = 1.2 \times 10^{23} \text{ m}^{-3}$

$a = 5.0 \text{ mm}$

$b = 0.20 \text{ mm}$

$I = 60 \text{ mA}$

$B = 0.080 \text{ T}$

Use this data to calculate

- (i) the mean drift velocity v of electrons within the semiconductor

$v = \text{-----} \text{ m s}^{-1} \text{ [3]}$

(ii) the potential difference V between the shaded faces of the slice.

$V = \text{-----} V$ [1]



EXAM PAPERS PRACTICE

- 10 Fig. 6.1 shows a single photomultiplier tube and its internal components. The tube can detect gamma photons in high-energy physics experiments.

A single gamma photon incident on the scintillator crystal generates many photons of blue light. These visible light photons travel to the photocathode where they are converted into photoelectrons. The number of electrons is then multiplied in the photomultiplier tube with the help of electrodes called dynodes. A short pulse of electric current is produced at the output end of the photomultiplier tube.

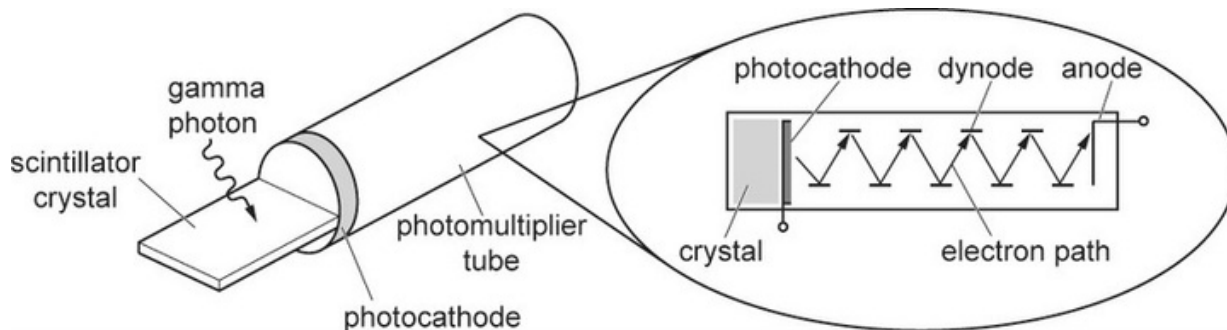


Fig. 6.1

The photocathode is coated with potassium which has a work function of 2.3 eV. Each emitted photoelectron is accelerated by a potential difference of 100 V between the photocathode and a metal plate, called the first dynode.

- (i) Show that the maximum kinetic energy of an emitted electron at the photocathode is very small compared to its kinetic energy of 100 eV at the first dynode.

- (ii) 2000 photoelectrons are released from the photocathode. Each photoelectron has enough energy to release four electrons from the first dynode at the collision. These four electrons are then accelerated to the next dynode where the process is repeated. There are 9 dynodes in the photomultiplier tube. The total number of electrons collected at the anode for each photoelectron is 49.

The pulse of electrons at the anode lasts for a time of 2.5×10^{-9} s.

Calculate the average current due to this pulse.

average current = A [3]

11 Which electrical quantity has S.I. units ampere-second (A s)?

- A charge current
- B resistance potential
- C difference
- D

Your answer

[1]



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12(a)

Derive the S.I. base units for resistance.

base units:----- [2]

(b) Fig. 16.1 shows the I - V characteristics of two electrical components L and R.

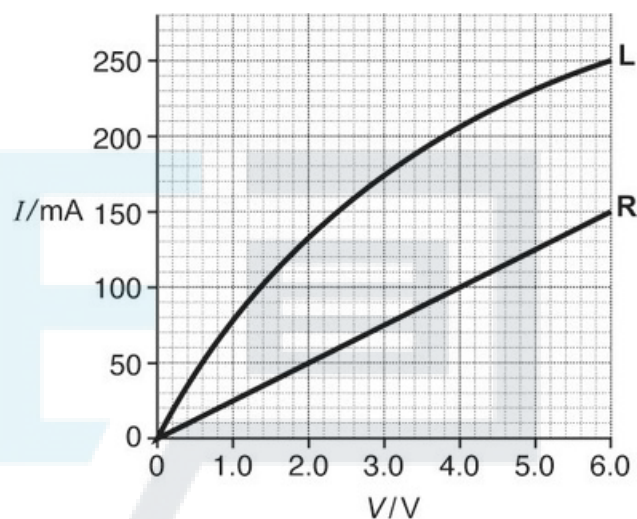


Fig. 16.1

The component L is a filament lamp and the component R is a resistor.

(i) Show that the resistance of R is $40\ \Omega$.

(ii) Fig. 16.2 shows the components L and R connected in series to a battery of e.m.f. 6.0 V.

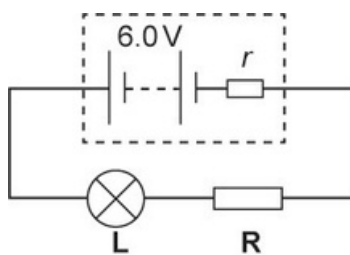


Fig. 16.2

The resistor R is a cylindrical rod of length 8.0 mm and cross-sectional area $2.4 \times 10^{-6} \text{ m}^2$. The current in the circuit is 100 mA.

- 1 Use Fig. 16.1 to determine the internal resistance r of the battery.



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$r = \dots\dots\dots \Omega$ [3]

- 2 Calculate the resistivity ρ of the material of the resistor R.

$\rho =$ ----- $\Omega \text{ m}$ [2]

- 3 There are 6.5×10^{17} charge carriers within the volume of R.

Calculate the mean drift velocity v of the charge carriers within the resistor R.



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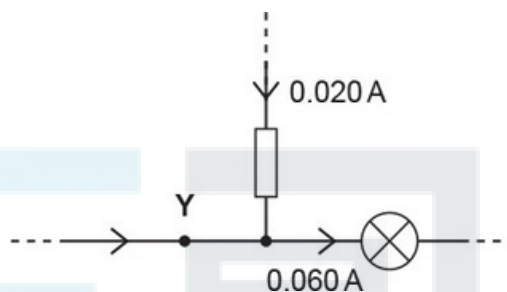
$v =$ ----- m s^{-1} [3]

- 13 Which law indicates that charge is conserved?
- A Lenz's law
- B Coulomb's law
- C Kirchhoff's first law
- D Faraday's law of electromagnetic induction

Your answer

[1]

- 14 Part of an electric circuit is shown below.



The direction of all the currents and the magnitude of two currents are shown.

How many electrons pass through the point Y in 10 s?

- A 1.25×10^{18}
- B 2.50×10^{18}
- C 3.75×10^{18}
- D 5.00×10^{18}

Your answer

[1]



- 15 Wires P and Q, made from the same metal, are connected in parallel across a cell of negligible internal resistance.

The table shows some data.

Wire	Length of wire	Diameter of wire	Mean drift velocity of electrons in the wire / mm s^{-1}
P	L	d	0.60
Q	$3L$	$2d$	v

What is the mean drift velocity v of the electrons in wire Q?

- A 0.15 mm s^{-1}
- B 0.20 mm s^{-1}
- C 0.30 mm s^{-1}
- C 0.60 mm s^{-1}

Your answer

[1]



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16(a) A 150 W heater constructed from nichrome wire is switched on for 5.0 hours.

The wire has a cross-sectional area of $4.1 \times 10^{-9} \text{ m}^2$.

The current in the wire is 1.5 A.

The number density of charge carriers in nichrome is $7.9 \times 10^{28} \text{ m}^{-3}$.

Calculate

(i) the resistance R of the heater

$R = \dots\dots\dots \Omega$ [2]

(ii) the number N of electrons passing through the heater in 5.0 hours

$N = \dots\dots\dots$ [2]

(iii) the mean drift velocity v of the electrons (charge carriers) in the heater.

$v = \dots\dots\dots \text{ms}^{-1}$ [2]

(b) Nichrome is a metal. Silicon is a semiconductor.

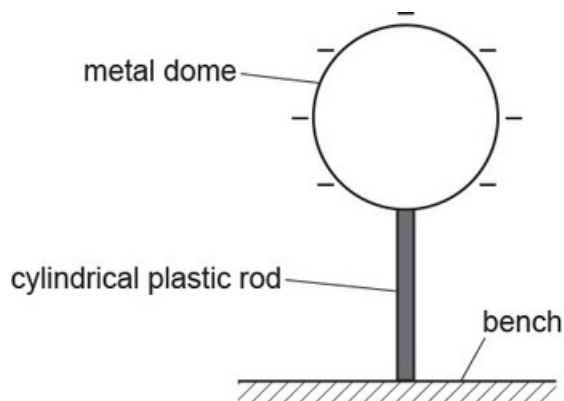
State how the number density of charge carriers n and the resistivity ρ of silicon compare with that of nichrome.

n

ρ

[2]

- 17 A spherical metal dome shown below is charged to a potential of -12 kV .



The dome is supported by a cylindrical plastic rod. The radius of the dome is 0.19 m .

- (i) Show that the magnitude of the total charge Q on the dome is $2.5 \times 10^{-7} \text{ C}$.

[2]

- (ii) The dome discharges slowly through the plastic rod.
It takes 78 hours for the dome to completely discharge.

- 1 Show that the mean current I in the plastic rod is about $9 \times 10^{-13} \text{ A}$.

- 2 The average potential difference across the plastic rod during discharge is 6000 V .
The rod has cross-sectional area $1.1 \times 10^{-4} \text{ m}^2$ and length 0.38 m .

[2]

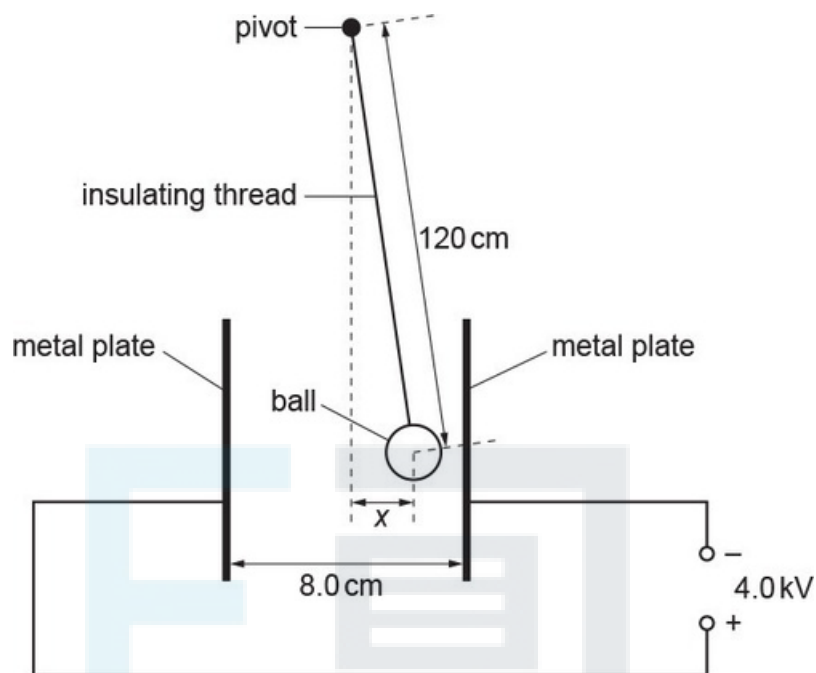
Calculate the resistivity ρ of the plastic.

$\rho = \dots\dots\dots \Omega\text{m}$ [3]

18(a) A ball coated with conducting paint has weight 0.030 N and radius 1.0 cm . The ball is suspended from an insulating thread. The distance between the pivot and the centre of the ball is 120 cm .

The ball is placed between two vertical metal plates. The separation between the plates is 8.0 cm . The plates are connected to a 4.0 kV power supply.

The ball receives a positive charge of 9.0 nC when it is made to touch the positive plate. It then repels from the positive plate and hangs in equilibrium at a displacement x from the vertical, as shown below. The diagram is not drawn to scale.



(i) Show that the electric force acting on the charged ball is $4.5 \times 10^{-4}\text{ N}$.

[2]

(ii) Draw, on the diagram above, arrows which represent the three forces acting on the ball. Label each arrow with the name of the force it represents.

[2]

(iii) By taking moments about the pivot, or otherwise, show that $x = 1.8$ cm.

[2]



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(b) The ball is still positively charged.

The plates are now moved slowly towards each other whilst still connected to the 4.0 kV power supply. The plates are stopped when the separation is 5.0 cm.

Explain the effect that this has on the deflection of the ball and explain why the ball eventually starts to oscillate between the plates.

[4]

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(c) When the ball oscillates between the plates, the current in the external circuit is $3.2 \times 10^{-8} \text{ A}$.

A charge of 9.0 nC moves across the gap between the plates each time the ball makes one complete oscillation.

Calculate the frequency f of the oscillations of the ball.

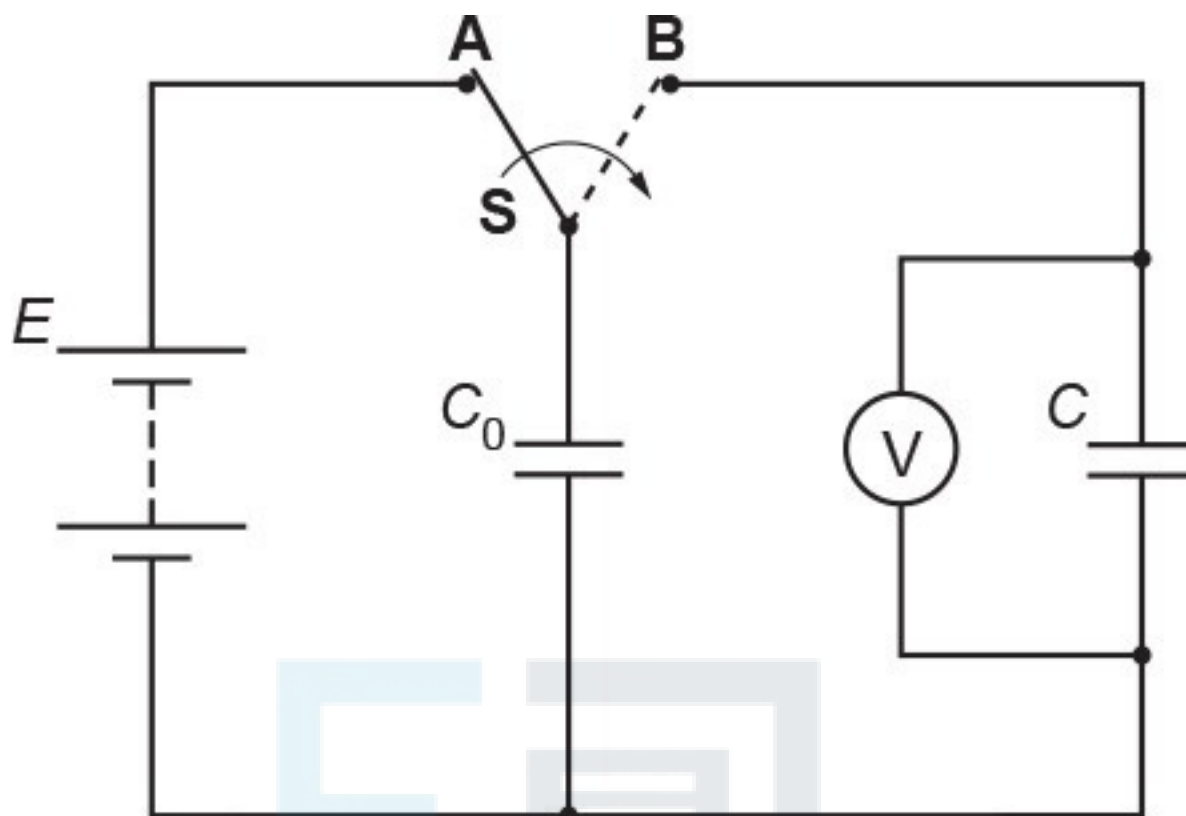
$f = \dots\dots\dots \text{ Hz [2]}$



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19(a) The diagram below shows a circuit containing two capacitors which are both initially uncharged. The battery has e.m.f. E and negligible internal resistance.

The switch S is first moved to position A until the capacitor of capacitance C_0 is fully charged.



The switch S is then moved to position B . The initial charge stored by the capacitor of capacitance C_0 is shared between the two capacitors.

The final reading on the voltmeter is V .

Show that $V = \frac{C_0}{C + C_0} E$.

[2]

(b) A student wants to determine the values of E and C_0 by repeating the experiment above and measuring the potential difference (p.d.) V for a selection of capacitors of capacitance C .

The student decides to plot a graph of $\frac{1}{V}$ against C .

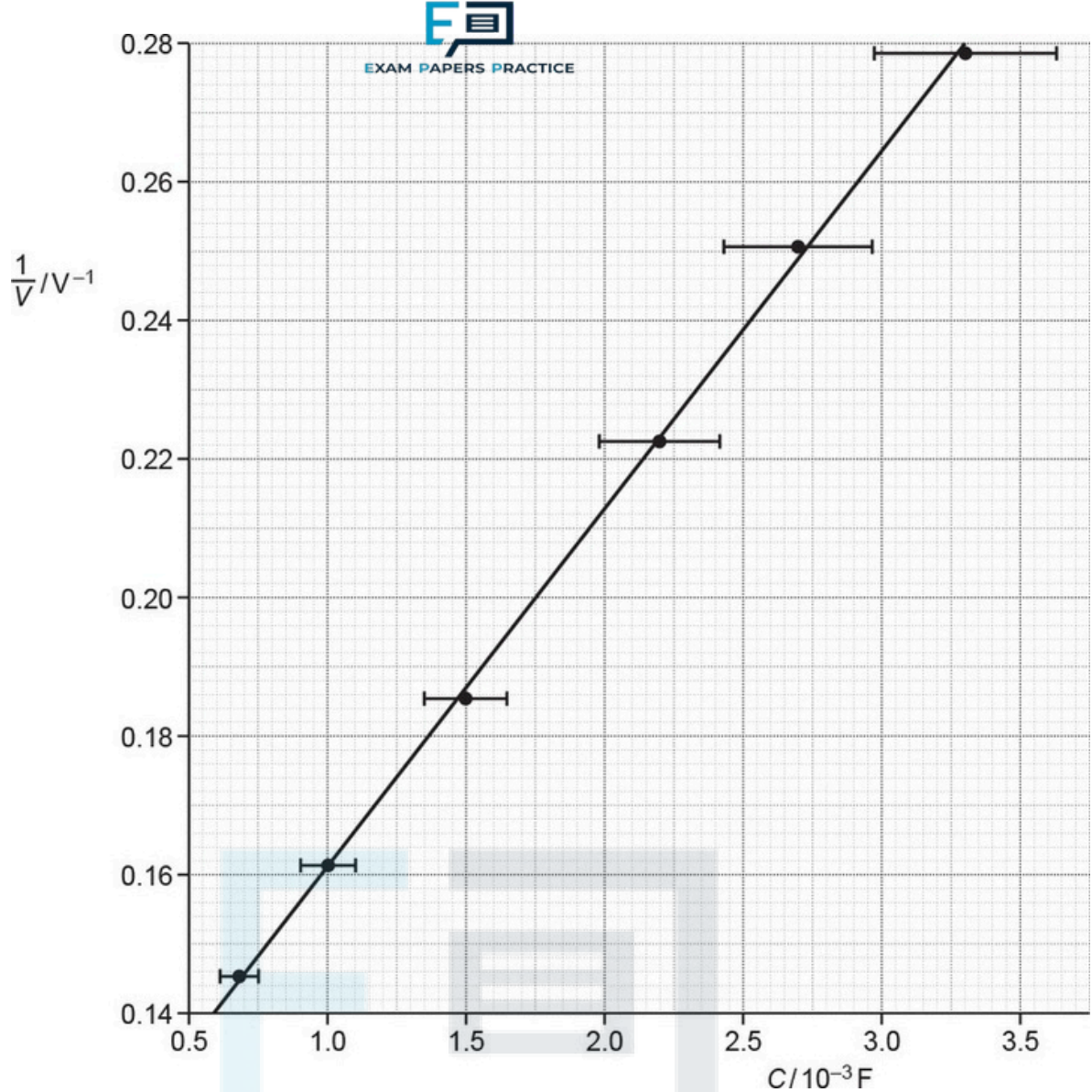
(i) Use the expression in (a) to show that the graph should be a straight line of gradient $\frac{1}{C_0 E}$ and y-intercept

$$\frac{1}{E}.$$

[1]

(ii) The data points, error bars and the line of best fit drawn by the student are shown in the graph below.

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The gradient of the line of best fit is $51 \text{ V}^{-1} \text{ F}^{-1}$. The value of E is 9.1 V .

Determine the value of C_0 in millifarads (mF). Write your answer to 2 significant figures.

$C_0 = \dots\dots\dots \text{ mF [2]}$

(iii) Draw on the graph a straight line of worst fit.

Use this line to determine the absolute uncertainty in your value of C_0 . Write your answer to an appropriate number of significant figures.

absolute uncertainty = mF [4]

- (c) The experiment is repeated with a resistor of resistance $10\text{ k}\Omega$ placed in series between S and the capacitor of capacitance C_0 .

State with a reason what effect, if any, this would have on the experiment.

[1]

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20 This question is about the Sun and its radiation.

* A student attends a lecture about the Sun and makes the following notes.

1. The Sun loses more than 4×10^9 kg of its mass every second to maintain its luminosity.
2. Treating hydrogen nuclei (protons) as an ideal gas, a temperature of 10^{10} K provides a kinetic energy of about 1 MeV, which is necessary for fusion.
3. However, the Sun's core temperature is only 107 K, so the chance of protons fusing on collision is very small. This explains why the Sun has such a long lifetime.

Explain the principles of physics which are involved in each of the three points.

You should include relevant formulae, but no numbers or calculations are required.

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[6]

21 This question is about investigations involving an electromagnetic wave.

A vertical transmitter aerial emits a vertically polarised electromagnetic wave which travels towards a vertical receiver aerial. The wavelength of the wave is 0.60 m.

Fig. 5.1 shows a short section of the oscillating electric field of the electromagnetic wave.

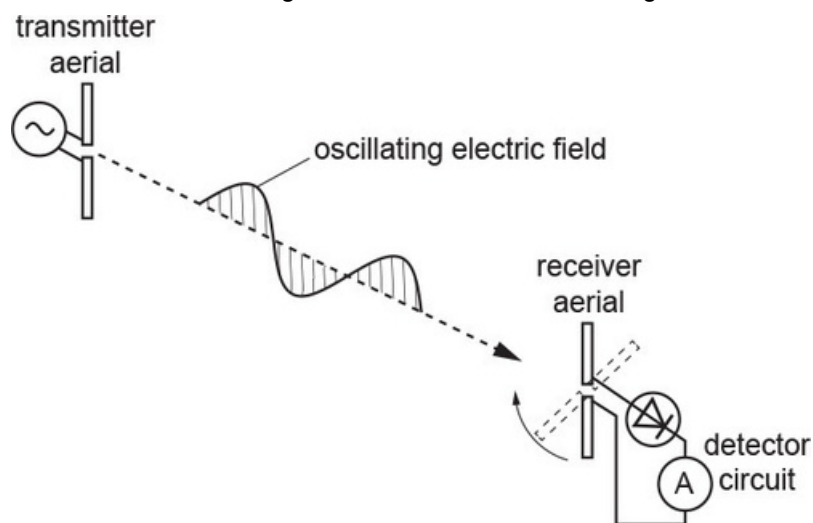


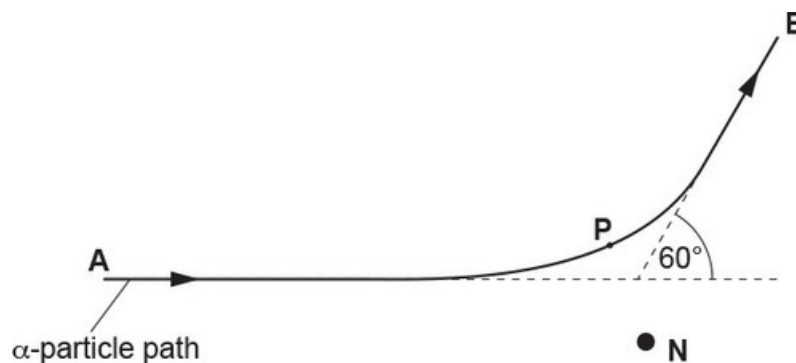
Fig. 5.1

Suggest why the diode in Fig. 5.1 is necessary for an ammeter to detect a signal at the receiver aerial.

[1]

- 22 A beam of α -particles is incident on a thin gold foil. Most α -particles pass straight through the foil. A few are deflected by gold nuclei.

The diagram shows the path of one α -particle which passes close to a gold nucleus N in the foil. The α -particle is deflected through an angle of 60° as it travels from A to B. P marks its position of closest approach to the gold nucleus.



The distance between P and N is 6.8×10^{-14} m.

Calculate the magnitude of the electrostatic force F between the α -particle (${}^4_2\text{He}$) and the gold nucleus (${}^{197}_{79}\text{Au}$)

when the α -particle is at P.

$F = \dots\dots\dots$ N [4]

23 Which sequence shows the materials arranged in the order of increasing number density of charge carriers?
increasing number density $\longrightarrow$

- A conductor, insulator, semiconductor
- B conductor, semiconductor, insulator
- C insulator, semiconductor, conductor
- D semiconductor, insulator, conductor

Your answer

[1]

END OF QUESTION PAPER



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